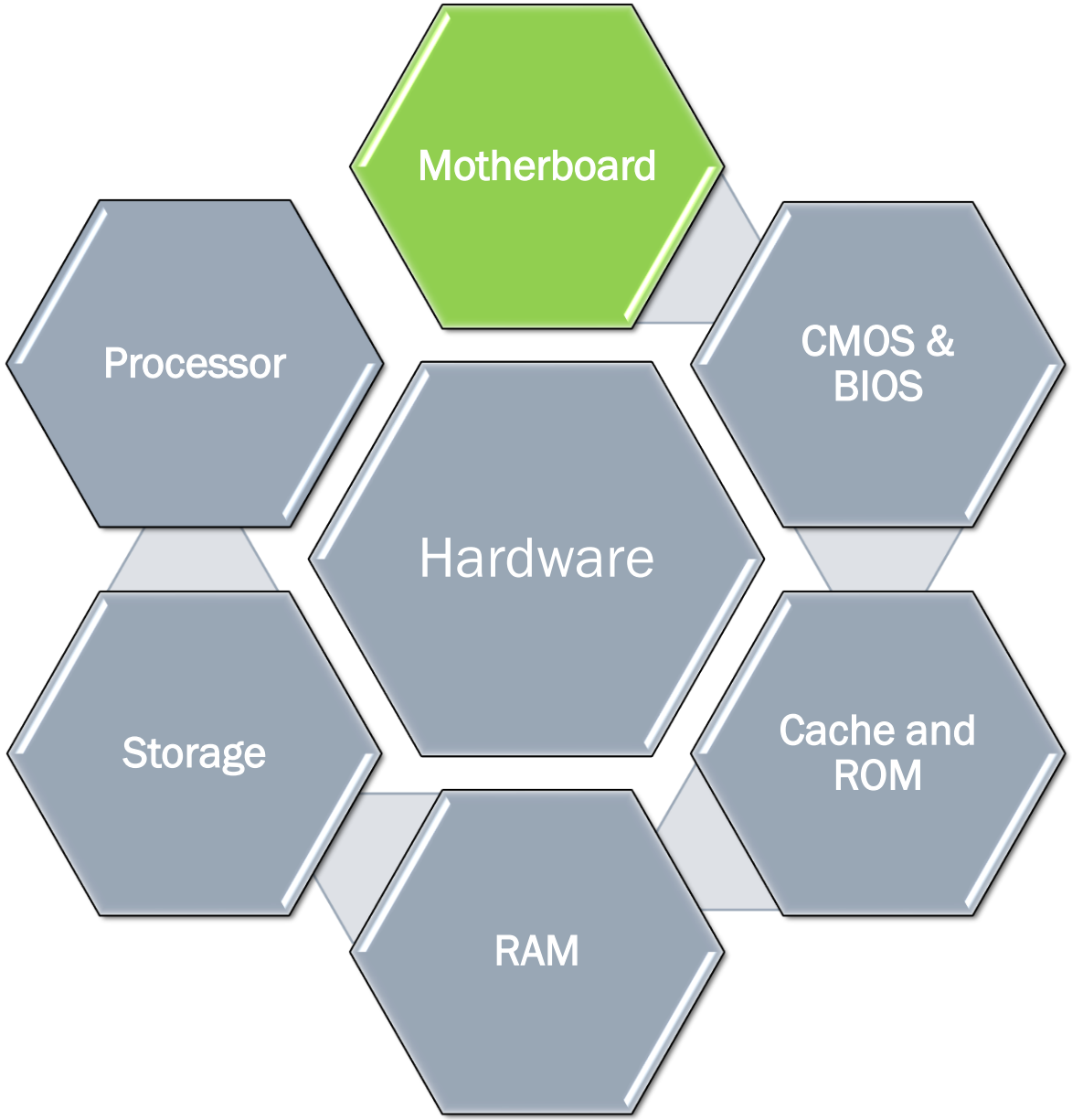
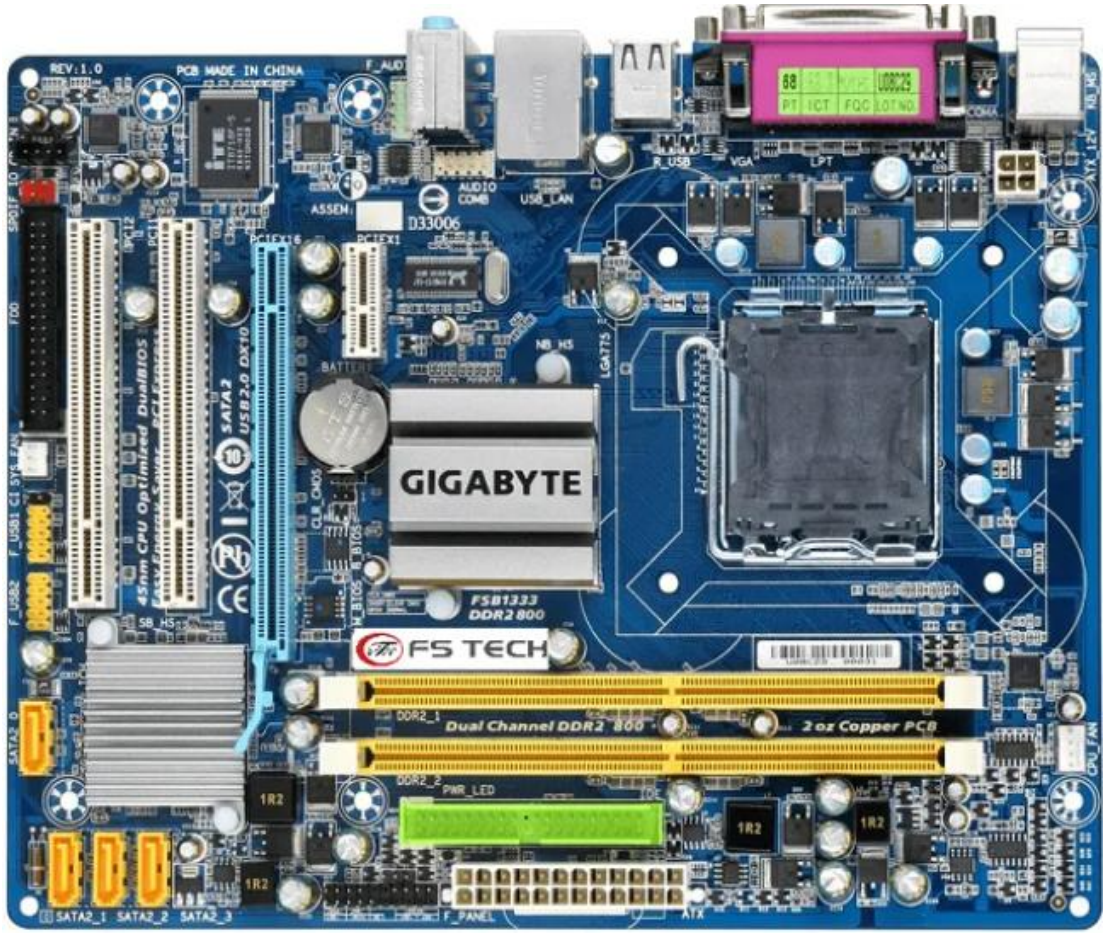
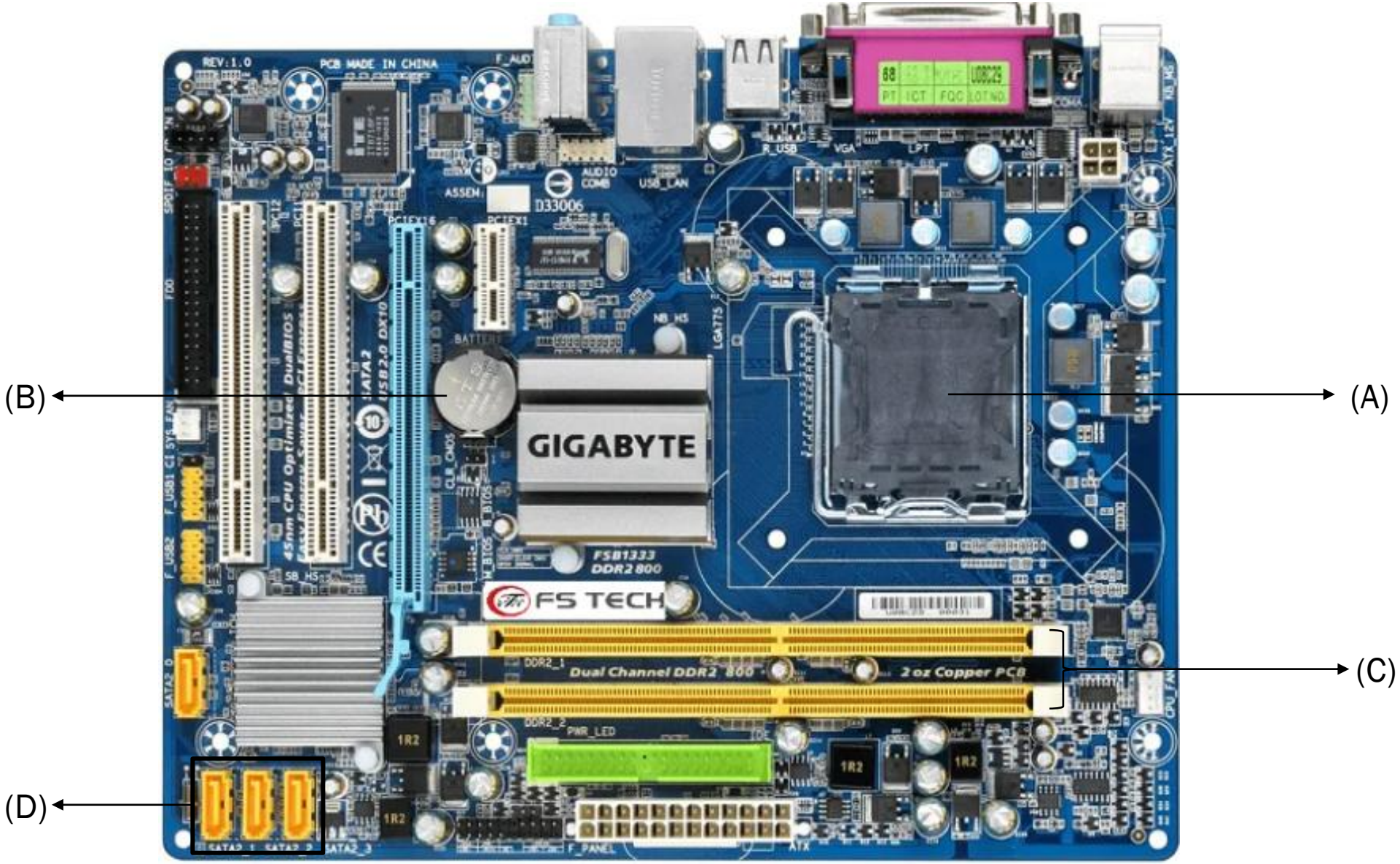

Motherboard
Processor
CMOS &
BIOS
Hardware
Storage
Cache and
ROM
RAM



1- The Motherboard

Overview

- The motherboard is the main circuit board that connects all parts of a computer together.
- Every component — **CPU, RAM, ROM, BIOS, CMOS, power connectors, storage**, etc. — is attached to it.
- It acts like the nervous system of the computer: everything communicates through it.

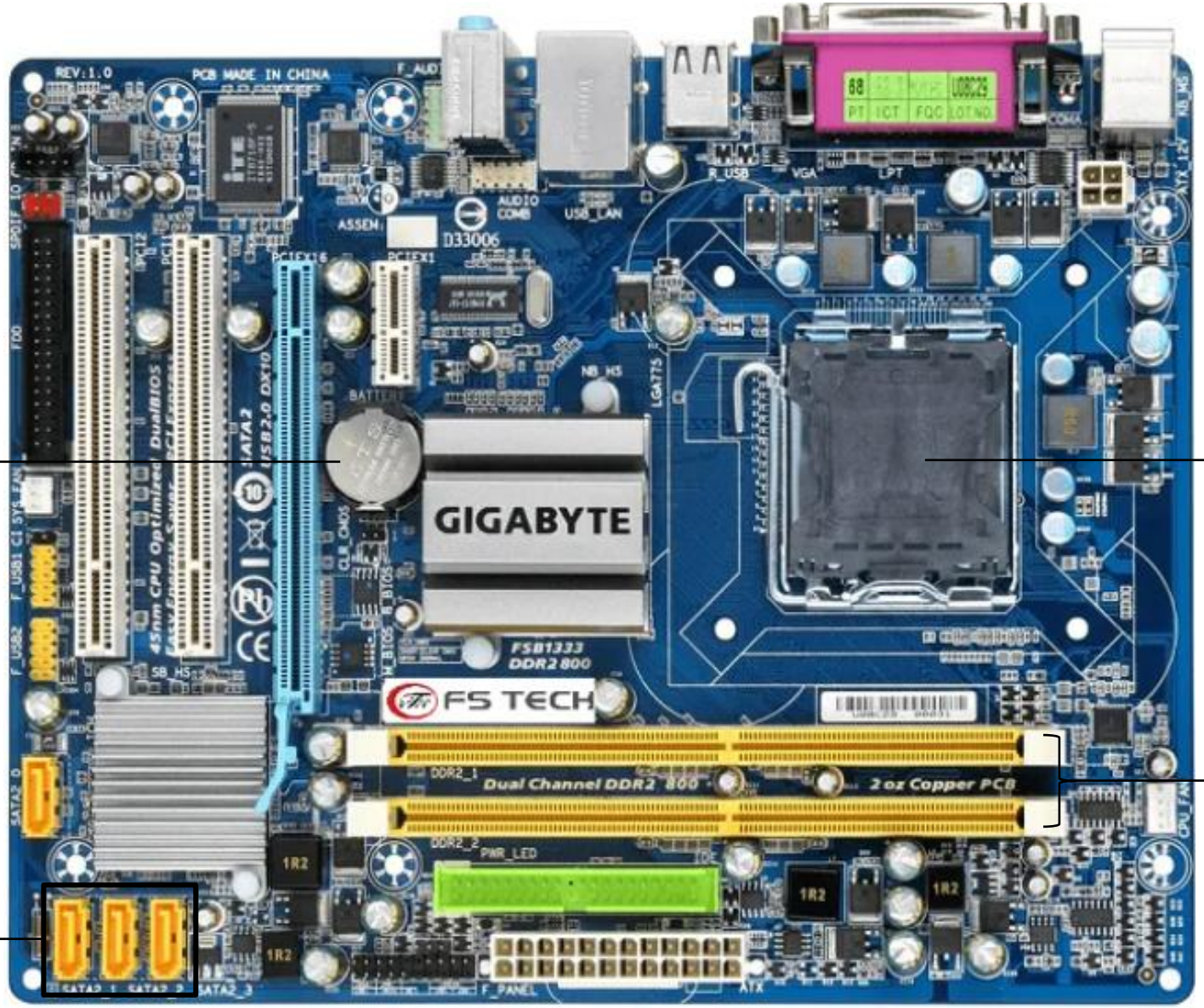

(B)
(A)
(C)
(D)
GIGABYTE
FS TECH
Dual Channel DDR2 800
2 oz Copper PCB
FSB1333
DDR2 800
PCB MADE IN CHINA
REV:1.0
PCIEX16
PCIEX1
LGA775
SATA2_1
SATA2_2
SATA2_3
F_PANEL
ATX

CMOS (B)

(A) CPU Socket

(C) RAM – Slots for memory

(D) Connectors for hard disk , USB drive and HDMI



Key Components

Component	Function	Example
CPU Socket	Holds the processor	Intel / AMD socket
Memory Slots (RAM)	Temporary data storage	8GB DDR4 RAM
Power Connectors	Supply power to board	24-pin ATX
ROM Chip	Stores firmware	BIOS program
BIOS & CMOS	Boot instructions + save settings	BIOS chip + battery

Example Analogy

Think of the motherboard as a **city**:

- **CPU** = Mayor
- **RAM** = Office space for short tasks
- **Storage** = City archive
- **Power** = Electricity grid
- **BIOS** = City startup plan

Section Quiz :

1- Accessing data from the hard disk is much faster than accessing it from the RAM

- True
- False

2- Which of the following instructs the computer on how to perform basic functions such as booting and keyboard control

- CMOS
- Processor
- BIOS
- RAM

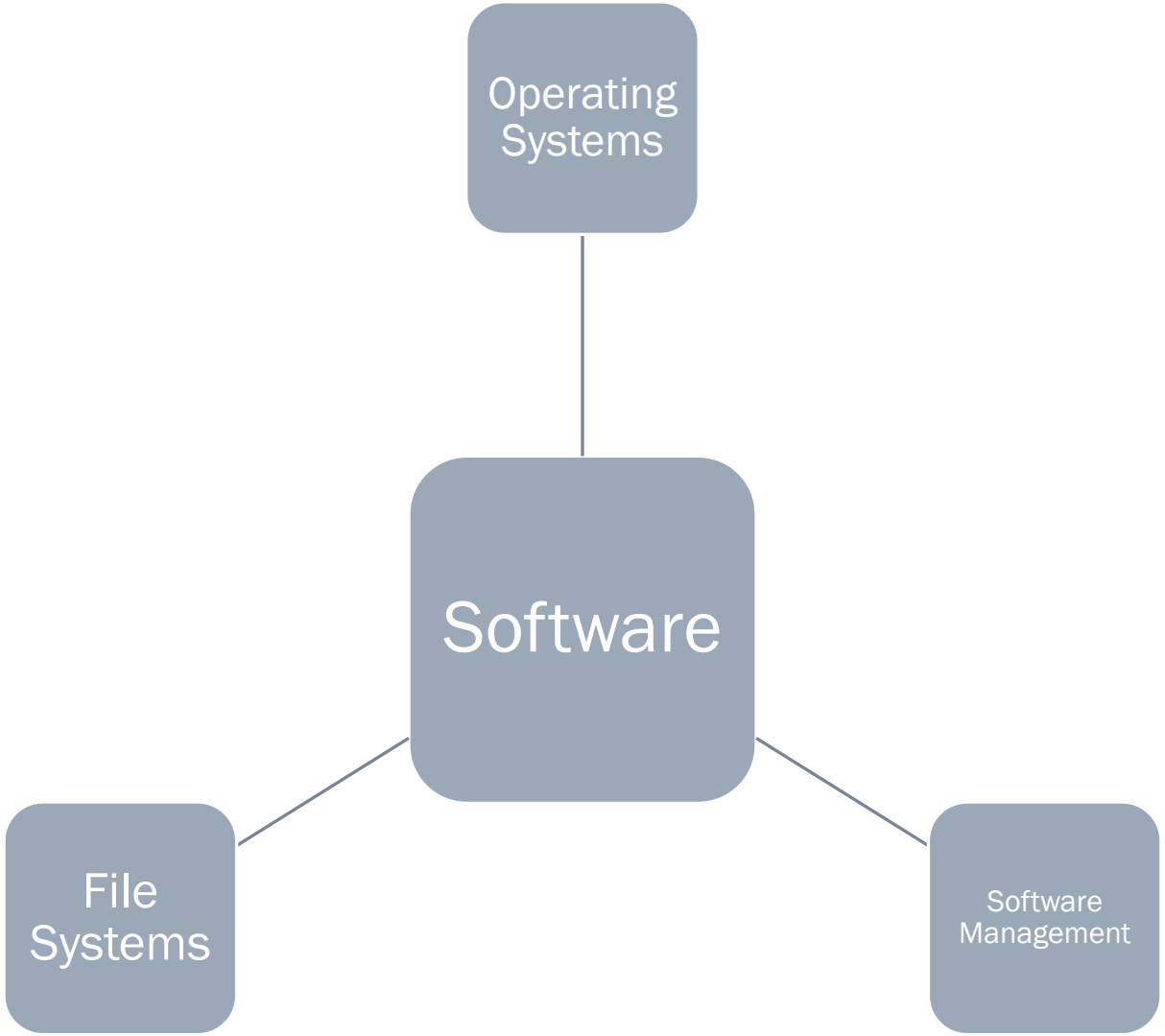
Section Quiz :

3- Data stored in the ROM is non-volatile and can only be read

- True
- False

4- Which of the following best describes the relationship between the BIOS and the CMOS?

- BIOS stores temporary data while CMOS controls input/output operations
- CMOS stores the BIOS program permanently
- BIOS contains the startup instructions, and CMOS saves the settings you change in the BIOS
- Both BIOS and CMOS are types of RAM


Operating Systems
Software
File Systems
Software Management

Software

What Is Software?

- Software = **A set of instructions** that tells the hardware what to do.
- Acts as a **bridge** between user and hardware.
- Example: You click “Print” → software sends command to printer hardware.

Types of Software

Type	Description	Examples
System Software	Controls and manages hardware. Platform for other software.	Windows, macOS, Linux
Firmware	Built-in software from manufacturer.	BIOS, router software
Application Software	Used for daily tasks.	Word, Excel, Photoshop
Utility Software	Maintains system health.	Antivirus, Disk Cleanup
Drivers	Help OS talk to hardware.	Printer driver, GPU driver

Driver Example:
When you plug in a printer — the **driver** helps your **Windows OS** communicate with it.

Utility Software

Category	Examples	Purpose
Security	Antivirus, Firewall, Encryption	Protect data and systems
Storage Management	Disk Defragmenter, Disk Analyzer	Organize data and optimize drives
Backup and Recovery	Cloud backup tools	Prevent data loss
Compression Tools	WinZip, 7-Zip	Reduce file size
Network Utilities	Ping, Traceroute	Monitor connectivity

Application Software

Type	Examples	Function
Office Software	MS Word, Google Docs	Documents, reports
Database Management Systems (DBMS)	MySQL, Oracle	Store and manage data
Computer-Aided Design (CAD)	AutoCAD, SolidWorks	Design products
Digital Graphics & Animation	Photoshop, Blender	Create visuals
ERP (Enterprise Resource Planning)	SAP, Odoo	Manage all business functions
Entertainment Software	Games, Media Players	Leisure and creativity
Communication Software	Games, Media Players	Messaging and collaboration

How Software Works

1. Software is written in **programming languages** (Python, C++, etc.).
2. A **compiler or interpreter** translates that code into **binary (0s & 1s)** that the CPU understands.

⚙️ **Example:**

- Compiler = checks the whole essay before reading it aloud.
- Interpreter = reads line-by-line and stops when there's an error.

1- Operating Systems (OS)

Operating Systems (OS)

What Is an OS?

- A **system software** that manages hardware and software.
- Provides a **Graphical User Interface (GUI)** — icons, windows, mouse, etc.
- Acts as a **middleman** between user and hardware.

⚙️ **Functions of an OS**

1. **Process Management** – Allocates CPU time to different applications and manages multitasking .
2. **Memory Management** – Allocates RAM efficiently to running programs .
3. **File System Management** – Organizes data storage using a file system (FAT32 , NTFS , ext4) storage.
4. **Device Management** – Handle drivers for hardware devices like printers , USBs , and graphics cards.
5. **Security & Access Control** – Manages user authentication , file permissions , and security protocols .
6. **User Interface** – Makes it easy to interact.

User Interfaces

Interface Type	Description	Example
Graphical User Interface (GUI)	Uses icons, windows, and menus	Windows, macOS
Command Line Interface (CLI)	Text commands typed by user	Terminal, CMD
Menu-Based Interface	Options are displayed in menus	ATMs, ticket machines
Adapted Interface	Customized for accessibility	Voice control, Braille display

Good interface design improves productivity, accessibility, and user satisfaction.

Types of OS

- Single-user vs. Multi-user OS

Type	Description	Example
Single-user	Designed for one user at a time	Windows 10, macOS
Multi-user	Supports multiple users together	Linux, Unix

Types of OS

- General – Purpose vs. Embedded OS

Type	Description	Example
General – Purpose OS	Used in personal Mobiles and Laptops	Android, iOS
Embedded OS	Custom operating system for a set of devices	ATM, Smart TV

Types of OS

- Desktop vs. Mobile OS

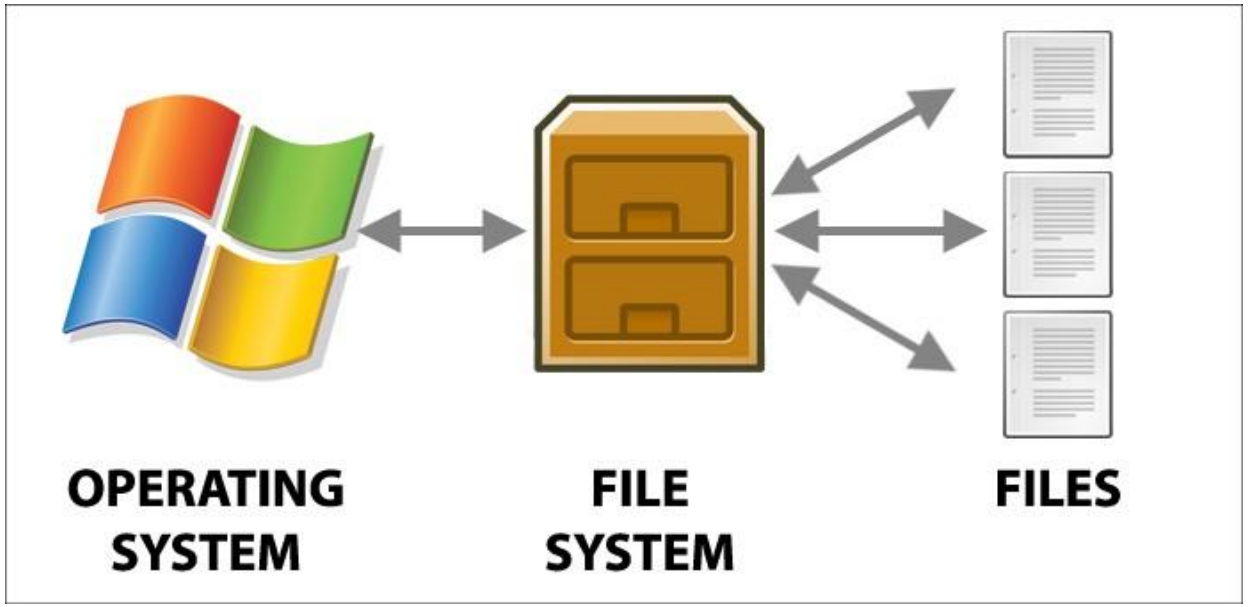
Type	Description	Example
Desktop OS	Runs on PCs and laptops	Windows , macOS , Linux
Mobile OS	Optimized for touch screens and mobile hardware	Android , iOS

Popular Operating System & Their Features

OS	Manufacturer	Main Features
Windows	Microsoft	User-friendly , supports most software , widely used for gaming and business
macOS	Apple	Secure , optimized for Apple devices , great for creative professionals.
Linux	Community-Driven	Open-source customizable , widely used for servers and development.
Android	Google	Open-source mobile OS , powers most smartphones.
iOS	Apple	Secure , optimized for Apple hardware , closed ecosystem.

Comparison (Summary)

Feature	Windows	macOS	Linux
User-Friendliness	High	High	Medium (depends on distribution)
Customization	Limited	Limited	High
Security	Medium	High	High
Software Availability	High	Medium	Medium
Performance	High	High	Varies by distribution



2- File Systems

File Systems

What Is a File System?

- A **method the OS uses to organize , store , and manage data** on storage devices, such as hard drives , SSD , and USB drives.
- It determines how data is structured and accessed.
- Works like a **library system** — everything has a label and location.

Key Functions

- Organizes data into files and directories (folders).
- Provides a method to store , retrieve , and modify files.
- Manages disk space efficiently.
- Ensures data security and access control.

Types of File Systems

- Different operating systems use different types of file systems , each with unique characteristics.

❑ Windows File Systems

- FAT32 (File Allocation Table 32)
  - Old but widely compatible.
  - Supports files up to 4GB in size
  - Used in USB drives and memory cards.

Windows File Systems

- **NTFS (New Technology File System)**
  - Default file system for Windows.
  - Supports files compression , encryption , and permissions.
  - Handles files larger than 4GB.

- **exFAT (Extended FAT)**
  - Designed for flash storage (USB drives , SD cards).
  - Supports large file sizes but lacks advanced security features.

❑ Apple File Systems

- **HFS+ (Hierarchical File System Plus)**
  - Used in macOS before 2017.
  - Supports journaling to prevent data corruption.

- **APFS (Apple File System)**
  - Modern , faster , and more efficient than HFS+.
  - Optimized for SSDs and flash storage.
  - Supports file cloning and encryption.

❑ Linux File Systems

- **ext3+ (Third Extended File System)**
  - Older but still used in some systems.
  - Supports journaling (prevent data loss).
- **ext4 (Fourth Extended File System)**
  - Default file system in most Linux distributions.
  - Faster and more reliable than ext3.
- **XFS, Btrfs, ZFS**
  - Advanced file systems used for enterprise storage.
  - Support high-performance operations and redundancy.

❑ Cross-Platform File Systems

- **exFAT** – Compatible with Windows , macOS , and Linux
- **FAT32** – Works on almost any system but has file size limitations.

File Naming Conventions

- **Windows:** Allows spaces, but some characters (/ \ * ? “ < > |)
- **Linux/macOS: Case-sensitive (**e.g. , File.txt and file.txt are different)
- **Extensions:** Used to identify file types (.txt , .jpg , .exe)

File Types and Formats

Category	Examples	Notes
Image	JPEG, PNG, GIF, SVG	Lossless or lossy compression
Video	MP4, AVI, MOV	Quality vs size trade-off
Audio	MP3, WAV, FLAC	MP3 = lossy, WAV = high quality
Documents	DOCX, PDF, TXT, RTF	Cross-platform sharing
Data/Software	XML, JSON, CSV	Structured data exchange

File System Operations

- **Create** – Generate a new file or directory
- **Read** – Open and view file contents.
- **Write/Modify** – Edit file contents.
- **Delete** – Remove files or directories.
- **Copy & Move** – Transfer files between locations.
- **Search** – Locate files using keywords or metadata.

3- Software Management

Software Management

- Software Installation

- **Local Installation** – The software is installed on a specific device (e.g. , Microsoft Office on a PC).
- **Network Installation** – Software is installed from a remote server onto multiple devices (e.g, enterprise applications).
- **Portable Software** – Does not require installation; runs from a USB or external drive (e,g,. Portable Chrome).
- **Cloud-Based Software (SaaS)** – No installation required , accessed via a web browser (e.g., Google Docs, Dropbox).

Software Updates & Patch Management

- **Manual Updates**– Users download and install updates themselves.
- **Automatic Updates**– Software updates in the background (e.g., Chrome auto-updates).
- **Security Patches:** Fix vulnerabilities to prevent cyber threats.
- **Feature Updates:** Add new capabilities and improve performance.

Open-Source Software

- **Free to Use:** Open-source software can be downloaded and used at no cost.
- **Source Code is Accessible**: Anyone can inspect, modify, or improve the code.
- **Community-Driven Development**: Maintained by a global community rather than a single company.
- **Highly Customizable**: Users can tailor the software to fit their needs.
- **Secure and Transparent**: Since the code is open, vulnerabilities can be identified and fixed faster.

Open-Source Software Example

Category	Software
Operating Systems	Linux (Ubuntu , Fedora), FreeBSD
Web Browsers	Mozilla Firefox , Brave
Office Suites	LibreOffice , OpenOffice
Programming Tools	Python, GCC , VS Code
Multimedia	VLC Media Player , GIMP
Database Management	MySQL

Open-Source Licensing Models

License	Software	Examples
GNU General Public License (GPL)	Modifications must be shared with the community	Linux, Wordpress
MIT License	Allows modification and distribution with minimal restrictions	React , Angular
Apache License	allow modifications with patent protection	Apache web Server , Android
BSD License	allows commercial use	FreeBSD , OpenSSH

Proprietary Software

- **Source Code is Not Public:** Only the company or developer has access to the code.
- **Usage Restriction:** Users must follow licensing terms, usually by purchasing a license.
- **Limited Customization:** Users cannot modify the software.
- **Official Support & Updates:** Companies provide technical support and regular updates.

Proprietary Software Example

Category	Software
Operating Systems	Microsoft Windows, macOS
Web Browsers	Google chrome , Safari
Office Suites	Microsoft office , Adobe Acrobat
Multimedia	Adobe Photoshop, Final cut Pro
Security Software	Norton Antivirus, Bitdefender
Cloud Services	Google Driver , Microsoft oneDrive

Key Differences Between Open-Source and Proprietary Software Example

Feature	Open – Source Software	Proprietary Software
Source Code	Available to the public	Closed , owned by a company
Customization	Users can modify & improve	No modification allowed
Cost	Usually free	Paid or subscription-based
Support	Community-driven support	Official customer support
Security	Transparent, issues fixed quickly	Relies on company to release patches
Examples	Linux, LibreOffice , Firefox	Windows, Microsoft office, Adobe Photoshop.

The Future of Open-Source vs. Proprietary Software

- **Hybrid Approaches:** Companies are blending both models (e.g., Microsoft now supports Linux)
- **Increased Open-Source Adoption:** Even major enterprises are embracing open-source software.
- **AI and Open Source:** AI development is benefiting from open-source models, with frameworks like TensorFlow and PyTorch.

Section Quiz :

1- What kind of software controls and manages hardware components?

- Application
- Utility
- Drivers
- System

2- Which of the following manages resources such as the memory and ensures smooth operation

- Operating system
- CPU
- Application software
- BIOS

3- Which of the following file systems are cross-platform compatible

- NTFS
- APFS
- FAT32
- ext4